MIL-C-44162
15 March 1985

MILITARY SPECIFICATION

CLOTH, PLAIN WEAVE, POLYESTER/COTTON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers blended polyester and cotton cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-Covered and Rayon Covered

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

Method 124 - Appearance of Durable Press Fabrics after Repeated Home Launderings

(Copies of Test Method may be obtained upon application to American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Materials.

3.2.1 Fibers.

3.2.1.1 Polyester. The fiber shall be polyethylene glycol terephthalate.

3.2.1.2 Cotton. The cotton shall be carded or combed.

3.2.2 Yarn. The warp and filling yarn shall be singles or 2-ply made from a blend of 60 to 85 percent polyester and the remaining percentage cotton, based on the dry weight of the desized cloth when tested as specified in 4.2.3.

3.3 Color. The color of the finished cloth shall be Black 464 and shall match the standard sample (see 6.3).

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3.3.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the cloth shall show no more than a slight trace of labile sulfur when tested as specified in 4.2.3.

3.2.2 Matching. The dyed and finished cloth shall match the standard sample under artificial daylight having a color temperature of 7000 ± 500 K, and shall be a good approximation to the standard sample under incandescent lamplight as 2850 ± 100K.

3.3.3 Colorfastness. The finished cloth shall show fastness to light, laundering, and crocking equal to or better than the standard sample. When no standard sample has been identified, the finished cloth shall show "good" fastness to light and laundering and shall have an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5 when tested as specified in 4.2.3.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.2.3.

TABLE I. Physical requirements

Weight ounces per sq yd minimum	Yarns per inch minimum		Breaking strength pounds, minimum		Tearing strength pounds, minimum	
	Warp	Filling	Warp	Filling	Warp	Filling
5.0	102	62	175	80	5.0	2.0

3.4.1 Weave. The weave shall be a plain weave.

3.4.2 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages when fly shuttle looms are used, and exclusive of selvages and fringe when shuttleless looms are used.

3.5 Cloth preparation. The cloth shall be desized, scoured, mercerized, and heat set prior to dyeing.

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3.6 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.2.3.

3.7 Dimensional stability. The finished cloth shall not shrink or elongate more than 2.0 percent in either the warp or filling direction when tested as specified in 4.2.3.

3.8 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.2.3.

3.9 pH. The pH value of the water extract of the finished cloth shall be no less than 5.5 nor more than 8.5 percent when tested as specified in 4.2.3.

3.10 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths each not less than 40 yards. The cloth shall be put-up on rolls as specified in 5.1.

3.11 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished cloth shall conform to the quality established by this document. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), E for slubs and B for knots, shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed 30.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 45.0 points. If one roll exceeds 45.0 points per 100 square yards, a second sample of the size indicated above shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

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$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Width less than minimum specified.
Poor dye penetration, mottled, streaky, or cloudy
Overall uncleanness

4.2.2.2 Length examination.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.2.2.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. Any roll off in the sample off shade; shaded side to side, side to center, or end to end; or not having the same appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for proper identification. The lot shall be unacceptable if two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

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4.2.3 End item testing. The cloth shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average result of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values used in expressing the final results. The sample unit shall be 4 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Polyester identification	3.2.1.1	1600 1/
Cotton identification	3.2.1.2	1200 1/
Fiber content	3.2.2	2535 1/
Labile sulfur	3.3.1	2020
Colorfastness to:		
Light	3.3.3	5660 2/
Laundering	3.3.3	5614
Crocking	3.3.3	5651
Weight	3.4	5041
Yarns per inch	3.4	5050
Breaking strength	3.4	5100
Tearing strength	3.4	ASTM D 1424
Weave	3.4.1	Visual 3/
Mercerization	3.5	1/
Nonfibrous material	3.6	2611
Dimensional stability	3.7	5556
Seam efficiency	3.8	5110 4/
pH	3.9	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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2/ The exposure time shall be 40 hours.

3/ One determination per sample unit and the result reported as "pass" or "fail".

4/ The needle shall measure 0.36 inch + 0.001 inch across the blade at the eye. The polyester/cotton wrapped thread shall conform to MIL-T-43548. The needle thread shall be 70, 2 ply and the looper thread shall be 80, 2 ply.

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth, shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required in the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth is intended for use as arm and shoulder patches in the fiber pile shirt of the Extended Cold Weather Clothing System.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Width of cloth required (see 3.4.2).
c. Length of roll required if other than specified (see 3.10).
d. Selection of applicable levels of put-up, preservation and packing (see 5.1 and 5.2).

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6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids (see 3.1).

6.4 Fabric defect scales. Fabric Defect Replica Kits are available from Sears Roebuck and Company, Department 817 (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684.

Custodians:

Army - GL
Navy - NU
Air Force - 11

Preparing activity:

Army - GL

Project No. 8305-0037

Review activities:

Army - MD
Navy - MC, CG
Air Force - 82, 99
DLA - CT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44162	Cloth, Plain Weave, Polyester/Cotton

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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